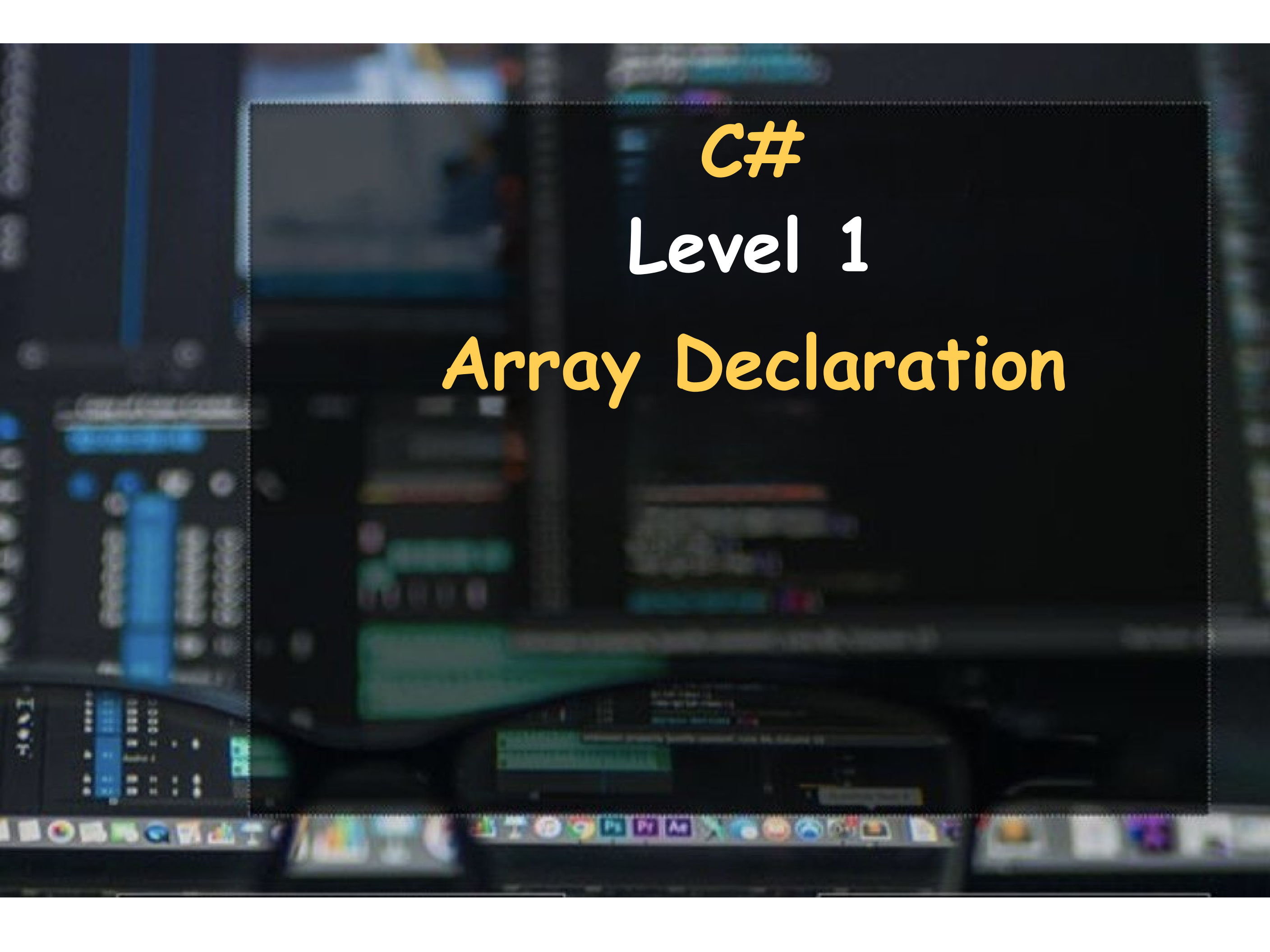




# C#

## Level 1

# Array Declaration



C#

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# Array Declaration

# Definition

## Array

-  An array is a collection of elements of the same data type under one variable name.
-  Array elements are accessed using indexes.

# Explanation

## Declaration

-  Declares an array without allocating its elements.

-  Syntax: `datatype[] arrayName;`

# 💡 Example

◆ `int[] age;`

## ▣ Memory Allocation

▨ The `new` keyword creates the array and specifies its length.

▨ `new int[5]` creates an array with 5 elements.

◆ `int[] age = new int[5];`

## ▣ Array Indexes

▨ Indexes start from 0.

▨ For length 5, valid indexes are  $0 \rightarrow 4$ .

## ▣ Accessing Elements

▨ Elements are accessed using their index.

`age[0] = 20;`

`age[1] = 25;`

`age[2] = 30;`

## Note

- ◆ Declaring an array and allocating it can be done separately or in one line.
- ◆ An array's length is fixed after it is created.
- ◆ Accessing an invalid index throws `IndexOutOfRangeException`.



# Comparison



## C#

- ✓ Array indexing is bounds-checked at runtime.
- ✓ Invalid indexes throw `IndexOutOfRangeException`.



## C++

- ✓ Built-in array indexing does not perform runtime bounds checking.
- ✓ Out-of-range access results in undefined behavior.

# ✦ Summary

- ✓ `datatype[] arrayName;` → declares an array.
- ✓ `new int[5]` → creates an array with 5 elements.
- ✓ Indexes start from 0.
- ✓ Last valid index = Length - 1.
- ✓ *C#* checks array boundaries at runtime.

Thank you 😊